

KHUSHAL HATKAR

khushalg@iisc.ac.in | [LinkedIn](#) | [Website](#) | +91 9768935069 | Indian Institute of Science, India

RESEARCH INTERESTS

Broadly interested in *in-situ* and *operando* electron microscopy and advanced characterisation to probe structure-property relationships in functional materials. My work centres on correlated oxides, but I am motivated by problems where nanoscale dynamics - phase transitions, domain evolution, defect-mediated phenomena - can be resolved through TEM, electron diffraction, and spectroscopy. I integrate computational tools (image simulation, 4D-STEM analysis, spectral processing) with experimental microscopy to extract quantitative insights.

EDUCATION

Indian Institute of Science (IISc)

BS-MS in Materials Science

Oct 2022 - Present

Bengaluru, India

- **Master of Science** (Expected Jul 2027)
- **Bachelor of Science (Research)** (Completed Apr 2026)
- **CGPA:** 8.3/10 (First Class) **Previous Semester TGPA:** 9.7/10
- **Focus:** Electron Microscopy, Raman Spectroscopy, Functional Materials, Characterisation, Opto-Thermal Mechanisms, Mechanical Behaviour

RESEARCH EXPERIENCE

Investigating *in-situ* heat-assisted phase transitions in In_2Se_3

Master's Thesis · Supervisor: Prof. Pavan Nukala

Aug 2026 - Present

Centre for Nanoscience Engineering
(CeNSE), IISc

- Using *in-situ* heating experiments inside the TEM to track heat-assisted phase transitions in $3\text{R In}_2\text{Se}_3$

Opto-Thermal Studies on VO_2 Microwires Using In-Situ Raman Spectroscopy

Bachelor's Thesis · Supervisor: Prof. Pavan Nukala

Aug 2025 - Present

Centre for Nanoscience Engineering
(CeNSE), IISc

- Investigating the modulation of the metal-insulator transition (MIT) temperature in VO_2 microwires through controlled nano-indentation, correlating strain with shifts in transition characteristics
- Performing *in-situ* opto-thermal experiments using **Raman spectroscopy** as both a probe and stimulus, mapping MIT behaviour as a function of laser power density and exposure duration across different temperatures
- Currently characterising nanoscale phase domain evolution during the MIT, aiming to resolve the monoclinic-rutile structural transformation at the domain level and also perform phase mapping using **TEM**
- **Tools & Software:** Python (SciPy, NumPy, Matplotlib, LMFIT), COMSOL Multiphysics, LabSpec 6, Fityk, Origin
- **Instruments:** Horiba LabRAM HR Raman Spectrometer, Park AFM NX20, ThermoFisher Tecnai T20-ST, ThermoFisher Titan Themis, Apreo 2S, Rigaku X-ray Diffractometer

Electro-Thermo-Mechanical Behaviour of Tin (Tin Metallography)

Research Project · Supervisor: Prof. Praveen Kumar

Aug 2024 - Jul 2025

Department of Materials Engineering,
IISc

- Performed **EBSD** orientation mapping to analyse texture evolution and grain boundary character distribution
- Conducted complementary **SEM**, **EDS**, and **WDS** analyses to understand composition and gain microstructural insights
- **Tools:** EBSD, EDS, WDS, SEM
- **Instruments:** Apreo 2S, ThermoFisher Helios G4-UX

INTERNSHIP EXPERIENCE

Summer Research Intern - Texture Analysis of Electrical Steels

Supervisor: Prof. S. Karthikeyan · Garrett Motion

May 2024 - Jul 2024

IISc

- Surveyed crystallographic texture analysis (ODF, pole figures, diffraction) and grain boundary engineering approaches to optimise Goss texture and minimise core losses in electrical steels

TEACHING EXPERIENCE

Teaching Assistant | Materials Design: Electronic, Electromechanical & Optical Functions (NE 240) | IISc *Aug 2026 - Dec 2026*

Physics & Chemistry Tutor | Tutored multiple high school students (grades 9-12) *Jan 2026 - Present*

TECHNICAL SKILLS

Microscopy: TEM (Tecnai T20 ST, Tecnai F30 - training in progress) · SEM · EBSD · EDS/WDS · AFM

Spectroscopy & Other: Raman Spectroscopy · X-Ray Diffraction (XRD)

Simulation & Analysis: COMSOL Multiphysics · CrystalMaker · VESTA · Origin · ImageJ · Fityk · LabSpec 6

Programming: Python (HyperSpy, abTEM, py4DSTEM, SciPy, NumPy, Matplotlib, pandas, scikit-image, LMFIT) · LaTeX/Typst · C · Java

Languages: English (Fluent, IELTS Band 8.0) · Hindi (Fluent) · Marathi (Native)

RELEVANT COURSEWORK

Advanced Electron Microscopy in Materials Characterisation (NE 316)	Electron Microscopy in Materials Characterisation (MR 306)
Applied Solid State Physics (NE 215)	Materials Thermodynamics (UMT 201)
Materials Design: Electronic, Electromechanical & Optical Functions (NE 240)	Functional Properties of Materials (UMT 309)
Material Synthesis: Quantum Dots to Bulk Crystals (NE 241)	Organic Electronics (MT 261)

CONFERENCES

2025 IEEE South Asia Ferroelectrics Symposium (SAFS) | Attendee *5-8 Aug 2025*

EMSI 2025 Conference | Attendee *12-14 Jul 2025*

AWARDS & FELLOWSHIPS

Kishore Vaigyanik Protsahan Yojana (KVPY-SA 2020) Fellow - All India Rank 297 | 2020-21
Fully funded federal scholarship program, Dept. of Science & Technology, Govt. of India

Microsoft Certified: Azure AI Fundamentals | Microsoft *Jun 2021*

EXTRACURRICULAR & LEADERSHIP

Convenor | Amalgam | UG Materials Science Club, IISc · organised Open Day 2026 *Jun 2025 - Mar 2026*
showcase for 50,000+ visitors

Core Team Member | Grain Boundary | Student-run club of the Materials Engineering Dept., IISc *Jun 2025 - Jun 2026*

Recreational Coordinator | Pravega XI | Core Committee of India's largest student-run SciTech-cultural fest, IISc *Jan 2025*

REFERENCES

Prof. Pavan Nukala CeNSE, IISc pnukala@iisc.ac.in	Prof. N. Ravishankar MRC, IISc nravi@iisc.ac.in	Prof. Praveen Kumar Materials Engg., IISc praveenk@iisc.ac.in	Prof. Abinandanan T.A. Materials Engg., IISc abinand@iisc.ac.in
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